MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 20: Residue Theorem Examples

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Lecture outline

1. Examples

Example 1

Find the residue of

$$f(z) = \frac{\cos z}{(z+i)^3}$$

at $z = -i$.

$$\mathrm{Res}(f, -i) = \lim_{z \to -i} \frac{1}{2!} \frac{d^2}{dz^2}((z+i)^3 \frac{\cos z}{(z+i)^3})) =$$

$$= \frac{1}{2} \lim_{z \to -i} (-\cos z) = \frac{-\cos i}{2} =$$

$$= -\frac{e^{i^2} + e^{-i^2}}{2} = -\frac{e^{-1} + e}{2} = -\frac{1}{2} \cosh 1$$

Example 2. Part 1.

Compute

$$\oint_\Gamma \frac{2iz - \cos z}{z^3 + z} dz$$

and Γ is any closed path that encloses $0, i$, and $-i$.

$$\oint_\Gamma \frac{2iz - \cos z}{z^3 + z} dz = \oint_\Gamma \frac{2iz - \cos z}{z(z^2 + 1)} dz = \oint_\Gamma \frac{2iz - \cos z}{z(z + i)(z - i)} dz =$$

$$= 2\pi i(\mathrm{Res}(f, 0) + \mathrm{Res}(f, i) + \mathrm{Res}(f, -i))$$

Example 2. Part 2.

$$\mathrm{Res}(f,0) = \lim_{z\to 0} \frac{2iz - \cos z}{(z+i)(z-i)} = -1$$

$$\mathrm{Res}(f,i) = \lim_{z\to i} \frac{2iz - \cos z}{z(z+i)} = \frac{-2 - \cos i}{-2} = 1 + \frac{1}{2}\cosh 1$$

$$\mathrm{Res}(f,-i) = \lim_{z\to -i} \frac{2iz - \cos z}{z(z-i)} = \frac{2 - \cos(-i)}{-2} = -1 + \frac{1}{2}\cosh 1$$

Thus

$$\oint_\Gamma \frac{2iz - \cos z}{z^3 + z} dz = 2\pi i \left(-1 + 1 + \frac{1}{2}\cosh 1 - 1 + \frac{1}{2}\cosh 1\right) = 2\pi i \left(-1 + \cosh 1\right)$$

Example 3. Part 1.

Let $f(z) = \dfrac{e^z}{(z+2i)^3 (z+i)}$. Evaluate

$$\oint_\Gamma f(z)\, dz$$

where Γ is any closed path enclosing $-i$ and $-2i$.

$$\oint_\Gamma f(z)\, dz = 2\pi i(\mathrm{Res}(f,-i) + \mathrm{Res}(f,-2i))$$

$$\mathrm{Res}(f,-i) = \lim_{z\to -i} \frac{e^z}{(z+2i)^3} = \frac{e^{-i}}{-i} =$$

$$= ie^{-i} = i(\cos(-1) + i\sin(-1)) = i\cos 1 + \sin 1$$

Example 3. Part 2.

$$\begin{aligned}
\operatorname{Res}(f,-2i) &= \lim_{z\to -2i} \frac{1}{2!}\frac{d^2}{dz^2}\frac{e^z}{z+i} \\
&= \frac{1}{2}\lim_{z\to -2i} \frac{e^z((z+i)^2-2(z+i-1))}{(z+i)^3} \\
&= \cos 2 - \frac{1}{2}\sin 2 + i\left(-\frac{1}{2}\cos 2 - \sin 2\right)
\end{aligned}$$

Finally, we get

$$\oint_\Gamma f(z)\,dz = 2\pi i\left(\sin 1 + \cos 2 - \frac{1}{2}\sin 2 + i\left(\cos 1 - \frac{1}{2}\cos 2 - \sin 2\right)\right)$$

Example 4. Part 1

Evaluate

$$\oint_\Gamma \frac{\sin z}{z^2\left(z^2+4\right)}dz$$

where Γ is a closed curve enclosing $0, 2i$, and $-2i$.

$$\oint_\Gamma \frac{\sin z}{z^2\left(z^2+4\right)}dz = \oint_\Gamma \frac{\sin z}{z^2(z+2i)(z-2i)}dz =$$

$$= 2\pi i\ \left(\mathrm{Res}(f,0) + \mathrm{Res}(f,2i) + \mathrm{Res}(f,-2i)\right)$$

$$\mathrm{Res}(f,0) = \lim_{z\to 0}\frac{1}{1!}\frac{d}{dz}\frac{\sin z}{z^2+4} = \lim_{z\to 0}\frac{d}{dz}\frac{\sin z}{z^2+4} =$$

$$= \lim_{z\to 0}\frac{(z^2+4)(\cos z)-(\sin z)(2z)}{(z^2+4)^2} = \frac{1}{4}$$

Example 4. Part 2

$$\mathrm{Res}(f, 2i) = \lim_{z \to 2i} \frac{\sin z}{z^2(z+2i)} = -\frac{1}{16}\sinh 2$$

$$\mathrm{Res}(f, -2i) = \lim_{z \to -2i} \frac{\sin z}{z^2(z-2i)} = -\frac{1}{16}\sinh 2$$

Finally, we get

$$\oint_\Gamma \frac{\sin z}{z^2\left(z^2+4\right)} dz = 2\pi i(\mathrm{Res}(f, 0) + \mathrm{Res}(f, 2i) + \mathrm{Res}(f, -2i)) =$$

$$= 2\pi i \left(\frac{1}{4} - \frac{1}{8}\sinh 2\right)$$

Example 5.

How many solutions does the equation

$$3z^6 - 14z^4 + 6z^2 - z + 1 = 0$$

have inside the circle $|z| = 2$?

This is equivalent to finding the number of roots that the polynomial $3z^6 - 14z^4 + 6z^2 - z$ has inside the circle $|z| = 2$.

Let $f(z) = -14z^4$ and $g(z) = 3z^6 + 6z^2 - z$

On $|z| = 2$ $|f(z)| = \left|-14z^4\right| = 14 \cdot 2^4 = 224$

$|g(z)| = \left|3z^6 + 6z^2 - z\right| \leq \left|3z^6\right| + \left|6z^2\right| + |-z| = 3 \cdot 2^6 + 6 \cdot 2^2 + 2 = 218$

Thus $|f| > |g|$ on the circle $|z| = 2$.

Since f has a zero of order 4, $f + g$ has 4 zeroes inside the circle.

Example 6. Part 1.

Show that

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3} = \frac{\pi^3}{32}.$$

Consider the series

$$\sum_{-\infty}^{\infty} \frac{(-1)^n}{(2n+1)^3}.$$

$$\sum_{-\infty}^{\infty} \frac{(-1)^n}{(2n+1)^3} = -\frac{\pi}{8}\mathrm{Res}\left(\frac{\csc \pi z}{\left(z-\frac{1}{2}\right)^3}, \frac{-1}{2}\right)$$

Example 6. Part 2.

$$\operatorname{Res}\left(\frac{\csc \pi z}{\left(z-\frac{1}{2}\right)^3}, \frac{-1}{2}\right) = \lim_{z \to -\frac{1}{2}} \frac{1}{2!} \frac{d^2}{dz^2} \csc \pi z$$

$$= \frac{\pi^2}{2} \lim_{z \to \frac{-1}{2}} \left(\csc \pi z \cot^2 \pi z + \csc^3 \pi z\right)$$

$$= -\frac{1}{2}\pi^2$$

Thus

$$\sum_{-\infty}^{\infty} \frac{(-1)^n}{(2n+1)^3} = \frac{\pi^3}{16}.$$

Example 6. Part 3.

But the series

$$\sum_{0}^{\infty} \frac{(-1)^n}{(2n+1)^3} = \sum_{-\infty}^{\infty} \frac{(-1)^n}{(2n+1)^3} - \sum_{-\infty}^{-1} \frac{(-1)^n}{(2n+1)^3}$$

$$= \frac{\pi^3}{2} - \sum_{-\infty}^{-1} \frac{(-1)^n}{(2n+1)^3} =$$

$$= \frac{\pi^3}{16} - \sum_{0}^{\infty} \frac{(-1)^n}{(2n+1)^3}$$

After solving this equation for $\sum_0^\infty \frac{(-1)^n}{(2n+1)^3}$ we get

$$\sum_{0}^{\infty} \frac{(-1)^n}{(2n+1)^3} = \frac{\pi^3}{32}$$

Example 7. Part 1.

Use the inversion theorem to find

$$(a)\quad L^{-1}\left[\frac{1}{\left(s^2-1\right)\left(s-1\right)^2}\right];$$

Since $L^{-1}\left[\phi\left(s\right)\right]=\sum$ residues of $e^{st}\phi\left(s\right)$ at each of its poles, we have

$$L^{-1}\left[\frac{1}{\left(s^2-1\right)\left(s-1\right)^2}\right]=\operatorname{Res}\left(\frac{e^{st}}{\left(s+1\right)\left(s-1\right)^3},1\right)+$$
$$+\operatorname{Res}\left(\frac{e^{st}}{\left(s+1\right)\left(s-1\right)^3},-1\right)$$

Example 7. Part 2.

$$\operatorname{Res}\left(\frac{e^{st}}{(s+1)(s-1)^3}, 1\right) = \frac{1}{2!}\lim_{s\to 1}\frac{d^2}{ds^2}\left(\frac{e^{st}}{(s+1)}\right) = \tfrac{1}{4}e^t t^2 - \tfrac{1}{4}e^t t + \tfrac{1}{8}e^t$$

$$\operatorname{Res}\left(\frac{e^{st}}{(s+1)(s-1)^3}, -1\right) = \lim_{s\to -1}\left((s+1)\frac{e^{st}}{(s+1)(s-1)^3}\right) = -\tfrac{1}{8}e^{-t}$$

Finally we get

$$L^{-1}\left[\frac{1}{(s^2-1)(s-1)^2}\right] = \frac{1}{4}e^t t^2 - \frac{1}{4}e^t t + \frac{1}{8}e^t - \frac{1}{8}e^{-t}$$

Example 7. Part 3.

Use the inversion theorem to find

$$(b)\ \ L^{-1}\left[\frac{1}{(s+a)^2+b^2}\right]$$

$$L^{-1}\left[\frac{1}{(s+a)^2+b^2}\right] = \operatorname{Res}\left(\frac{1}{(s+a)^2+b^2}, -a-bi\right) + \\ + \operatorname{Res}\left(\frac{1}{(s+a)^2+b^2}, -a+bi\right)$$

$$\operatorname{Res}\left(\frac{1}{(s+a)^2+b^2}, -a-bi\right) = \lim_{s\to -a-bi}\left(\frac{e^{st}}{s+a-bi}\right) = \\ = \frac{1}{2b}e^{-ta}\sin tb + \frac{1}{2}\frac{i}{b}e^{-ta}\cos tb$$

Example 7. Part 4.

$$\operatorname{Res}\left(\frac{1}{(s+a)^2+b^2}, -a+bi\right) = \lim_{s\to -a+bi}\left(\frac{e^{st}}{s+a+bi}\right) =$$

$$= \frac{1}{2b}e^{-ta}\sin tb - \frac{1}{2}\frac{i}{b}e^{-ta}\cos tb$$

Finally we get

$$L^{-1}\left[\frac{1}{(s+a)^2+b^2}\right] = \frac{1}{b}e^{-ta}\sin tb$$

Thank you